

BOLTZMAN CONSTANT KIT®

OBJECTIVE

MEASUREMENT of Boltzman's Constant using semiconductor Diode.

Conventional method to determine Boltzman Constant (k) make use of the block body Radiation and its two famous laws i.e. Wien's displacement law and Stefan's law. These methods are time consuming and very often lead to comparatively less accurate results.



The V – I characteristics of a semiconductor diode can be used to determine Boltzman Constant accurately and with simple equipment that can be handled with ease and convenience.

Principle: The diode equation is given by:

$$I = I_0 \left\{ \exp \left(\frac{eV}{\eta kT} \right) - 1 \right\} \quad \dots(1)$$

where V = voltage across the diode

I = forward current at voltage V

I_0 = reverse saturation current

k = Boltzmann Constant

T = Temperature in Kelvin

η = a constant, characteristic of the material from which the diode is made, for Ge diode, $\eta = 1$ while $\eta = 2$ for Si diode.

If $V \gg kT/e$, the Boltzmann constant can be expressed as

$$k = \frac{e}{\eta T \left(\frac{\Delta \ln I}{\Delta V} \right)} \quad \dots\dots\dots (2)$$

where, $\left(\frac{\Delta \ln I}{\Delta V} \right)$ is the slope of the straight line drawn between V and $\ln(I)$.

INSTRUMENT

The experimental kit provides the following facilities :

- A digital millivoltmeter (0 – 9.99 V) to measure the voltage across the diode.
- A highly stabilized power supply whose voltage can be varied in steps of 1 mV using a ten – turn potentiometer.
- A digital current ammeter (0 – 500 mA).
- Diodes: Silicon diodes & Germanium diodes.

Manufacturers:



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